



Type ZWC De-energized Tap-changer Operating Instructions

Shanghai Huaming Power Equipment Co.,Ltd.

Please read this instruction carefully before using the purchased tap-changer. Pay attention to the following items:

1. When receiving the product, check it according to the packing list. Evidences should be retained for damages caused by transportation in order to claim compensation from the responsible party and to protect your rights.
2. Every purchased tap-changer can be used only on the transformer specified in the order sheet. In case of changing the tap-changer application, please consult with Huaming first.
3. Product installation, commissioning, maintenance and repairment shall be in accordance with this operating manual and relevant safety regulations.

Important safety instructions are emphasized in this manual in the following four ways:



The "Warning!" sign is given, when neglecting a request may cause injuries to operators. It indicates that life and health are at risk. Neglecting this warning can lead to serious or fatal injuries.



The "Careful!" sign is given when neglecting a request may cause damages to equipment. It indicates that this product or user's other device is at risk, including serious or fatal injuries.



The "Attention" sign is given for the purpose of emphasizing. It requires users to be extremely cautious when dealing with requests with the "Warning" and "Careful" signs.

Reminder

This is a supplementary explanation for a specific matter.

Contents

1. General Information	1
2. Technical Specifications	2
3. Type Explanation	3
4. Connection Diagram	3
5. Application and Service Condition	4
6. Structure Features	4
7. Product Acceptance	5
7.1 Scope of delivery	5
7.2 Acceptance and storage	5
8. Installation	5
8.1 Tap-changer installation	6
8.2 Assembly of drive shaft and motor drive unit	6
8.2.1 Drive shafts assembly between two tap-changers	9
8.2.2 Assembly of the drive shaft between the tap-changer and the drive mechanism ..	11
9. Transformer Ratio and DC Resistance Test	15
10. Drying and Oil Filling	15
10.1 Vacuum drying	15
10.2 Vapor phase drying	15
11. Operation and Maintenance	16
11.1 Tap-changer switching operation and description	16
11.2 Maintenance	16
12.1 Leakage	17
12.2 Abnormal DC resistance	17
13. Warranty	18
14. Appendix	19
14.1 Tap-changer pictures	19
14.2 Drive mechanism picture	20
14.3 Assembly of drive mechanism and drive shafts	20
14.4 Dimension drawing with shielding	21
14.5 Dimension drawing without shielding	22
14.6 Assembly of tap-changer	23



Photos, charts and other data in this manual may differ from tap-changers actually delivered.

These drawings are for reference only and we reserve the right to make changes without notice.

1. General Information

This ZWC de-energized tap-changer (hereinafter referred to as tap-changer) is suitable for oil-immersed power transformers and special transformers, such as furnace or rectifier transformer, with a rated frequency of 50 or 60 Hz, a maximum equipment voltage of 145kV and a maximum rated current of 1000A. Tap changing takes place when the transformer is de-energized, so as to change the transformer ratio and stabilize the output voltage.

ZWC tap-changer complies with IEC 60214-1: 2014 Tap Changers – Part 1: Performance requirement and test methods and IEEE Std C57.131-2012 Requirements for Load Tap Changers.



Each tap-changer can only be used on the transformer specified in the tap-changer order specification.

The tap-changer installation, electrical wiring and commissioning must be carried out by qualified and skilled personnel in accordance with this manual.

It is not allowed to change or modify the tap-changer without prior consultation with Huaming.

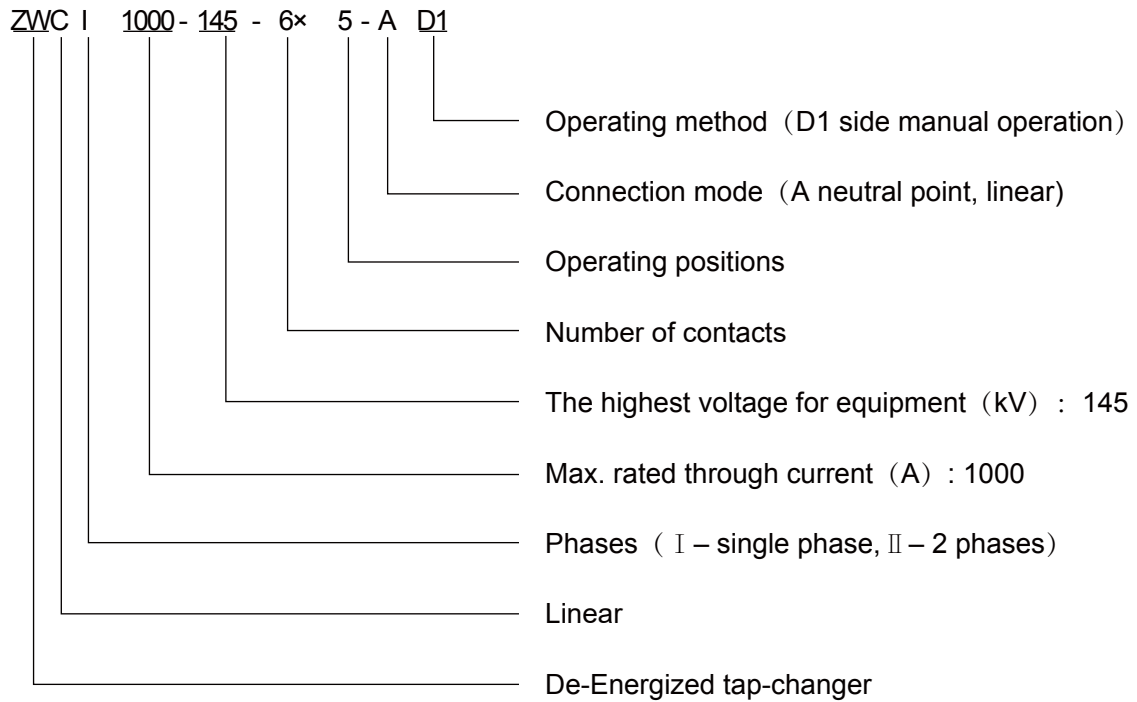
During tap-changer installation, electrical wiring and commissioning, failure to operate in accordance with this manual may cause malfunctions of motor drive units, tap-changers and transformers or even personal accidents and equipment damages.

2. Technical Specifications

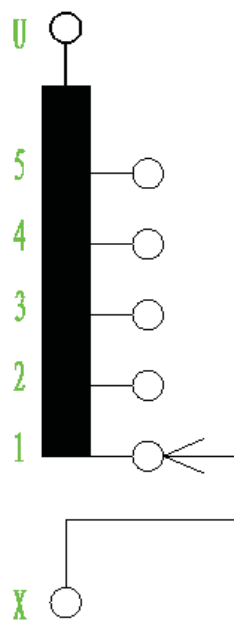
1	Tap-changer Model		ZWCI 500	ZWCI 1000	
2	Phases		Single phase		
3	Rated frequency（Hz）		50/60		
4	Rated through current（A）		500	1000	
5	Short-circuit current test（kA）	Thermal（3s）	8	15	
		Dynamic (Peak)	12.5	25	
6	Insulation level	Voltage（kV）		138	138
		The highest voltage for equipment（kV）		145	145
		Rated separate source AC withstand voltage（1min）（kV）	To ground or phase	275	275
			Between contact	50 with 50mm spacing	50 with 50mm spacing
				80 with 80mm spacing	80 with 80mm spacing
			Between max. & min. taps of the same	100	100
		Rated lightning impulse withstand voltage（1.2/50μs）（kV）	To ground or phase	650	650
			Between contact	140 with 50mm spacing	140 with 50mm spacing
				225 with 80mm spacing	225 with 80mm spacing
			Between max. & min. taps of the same phase	250	250
		7	Drying temperature（℃）		Vacuum drying 110℃, Vapor phase drying 125℃
8	Connection mode		Linear		
9	Operating method		Side manual operation		
10	Max. operating positions		5		
11	Contact resistance between fixed and moving		≤350μΩ		
12	Partial discharge（pC）		≤50		
13	Leakage Test		No leakage under 6x10 ⁴ PA for 24h		
14	Mechanical life		More than 10,000 operations		
15	Temperature rise of contact at 120% of rated		Below 15K (in oil)		

Note: Users should consider the insulation distances of tap-changer to the ground, to coils and other accessories. Enough space must be offered during design.

3. Type Explanation



4. Connection Diagram



Connection Diagram

5. Application and Service Condition

- 5.1 The temperature of the transformer oil shall be less than 105°C and higher than -25°C.
- 5.2 The tap-changer should be stored in air with ambient temperature of -25°C to 40°C and with relative humidity no more than 85%.
- 5.3 The tap-changer shall be stored in a safe area without corrosive gas.
- 5.4 The tap-changer should be used as soon as possible. When stored, do not remove the packing.
- 5.5 If the application conditions do not meet the above requirement, please contact our technical department for confirmation.

6. Structure Features

- 6.1 The structure consists of two parts: the tap-changer body and the drive mechanism. The transformer manufacturer shall connect the tap-changer with the drive mechanism.
- 6.2 The tap-changer is of linear type with a frame made of two insulating and supporting plates. It is intended to be used in power transformers and large-capacity distribution transformers. It is installed in the space between the windings and the tank wall. It benefits from a compact size and reduced tapping lead.
- 6.3 The tap-changer step-by-step operation is controlled by a Geneva wheel to ensure accurate positioning between the moving and the fixed contacts.
- 6.4 The appropriate connection between the tap-changer contacts is done through a classic clamping-type floating point method to meet the current carrying capacity.
- 6.5 When necessary the tap-changer can be equipped with a shielding structure. This provides a uniform electric field is uniform and good partial discharge performance
- 6.6 The tap-changer is provided with a positioning pin to avoid changes of tap position during installation so as to ensure correct installation and product reliability.
- 6.7 The tap-changer manual drive mechanism is designed for side operation. During one operation, the operating gear will rotate by one complete cycle. The handle of the drive mechanism is fixed on its end cap and can be used as an end limit and a locking component.

6.8 The drive shaft is suitable for installation on both drum and bell-type transformers. It has a telescopic drive shaft with a telescopic distance of 200mm allowing for simple installation.

7. Product Acceptance

7.1 Scope of delivery

Once the tap-changer and the drive mechanism have completed the routine tests, they are set to the middle tap position and then protected from humidity. The complete tap-changer assembly are shipped as separate assembly made of:

- ① ZWC de-energized tap-changer
- ② Drive mechanism
- ③ Drive shafts

The products should be carefully handled during transportation, protected from rain and moisture, and must be stored in upside position.

7.2 Acceptance and storage

When receiving the product, confirm the part received are in accordance with the packing list. Evidences should be retained of damages caused by transportation in order to claim compensation from the responsible party and to protect your rights.

Store the tap-changer in a warehouse where the air is unobstructed, the relative humidity is less than 85%, and the temperature is between -25 °C and + 40 °C. The storage environment shall not have corrosive gas and shall not be affected by rain or snow.

The tap-changer is stored in an airtight enclosure that shall only be open for installation.



Warning

When working on the tap-changer, drive mechanism and other components, all components must be securely fastened. Otherwise, there may be a risk of falling part, causing damages or serious injury .

8. Installation

Before installation:

1. Before installation, check whether the tap-changer specification and contract are in line with the transformer number and design requirements; check the serial numbers and specifications of the tap-changer, drive mechanism and drive shafts are consistent; check whether the components are

defective. Install the tap-changer and drive mechanism according to the directions on the approved drawing, especially when multiple tap-changers operate in parallel.

2. Due to the influence of tap-changer transportation, storage and drying environment, after being dried, some tap-changers' silver-plated contacts may appear yellow. This discoloration will not affect tap-changers' performance. In severe cases, rubber can be used to clean it.

8.1 Tap-changer installation

The tap-changer should be attached on the transformer according to design requirements. The tapping position terminals should be reliably and firmly connected to the transformer coil tapping leads according to the marks.



1. The length of the tapping cables should be appropriate. When the cables are being fixed, ensure that they are long enough to prevent the tap-changer from being pulled. At the same time, it is forbidden to tap or forcibly pull the cables or tap-changer. This could result in deformation of the tap-changer which can affect mechanical and electrical performance.

2. The tap-changer should be fixed firmly to prevent it from moving. Otherwise, the drive shaft may be disengaged from the tap-changer or subjected to radial shear force leading to damage. The space between fixations on the fixed tap-changer support mounting plate should be appropriate. The recommended error is $\pm 1\text{mm}$ and the size of the spacer should be consistent. If variation between spacers is too large, the tap-changer support plate will be deformed, especially when it's dry. It may also cause the plate to break.

3. If necessary, add insulating paper or cardboard between the tap-changer, the coil and the tank to further ensure the insulation level.

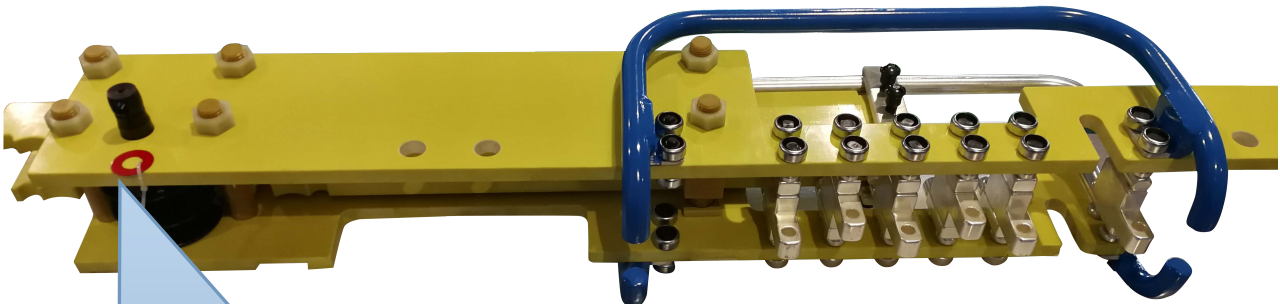
8.2 Assembly of drive shaft and motor drive unit

Before assembly:

1. Before the drive shaft is installed and connected, please disassemble the relevant positioning part in advance. The installation process must be based on the installation steps in the manual. (as is shown below)

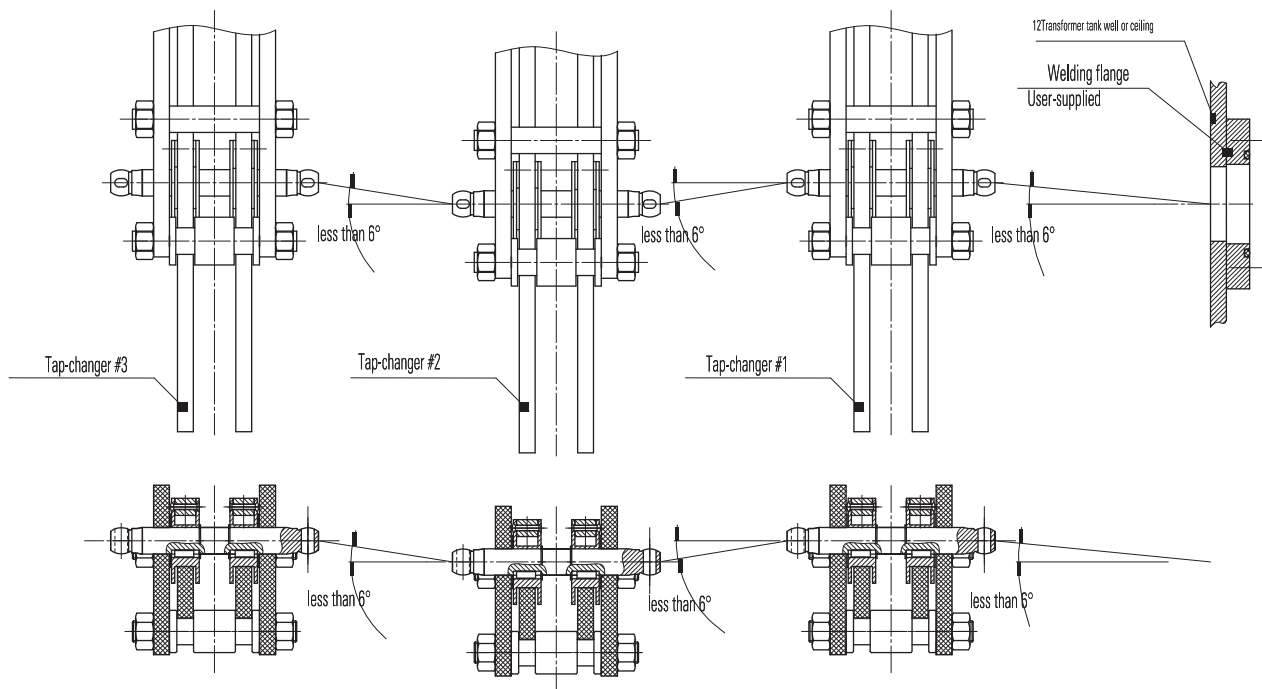


The approved DC resistance of the transformer cannot prove that the tap-changer is installed and operated correctly. The installation and verification must be in strict accordance with the requirements of the manual.

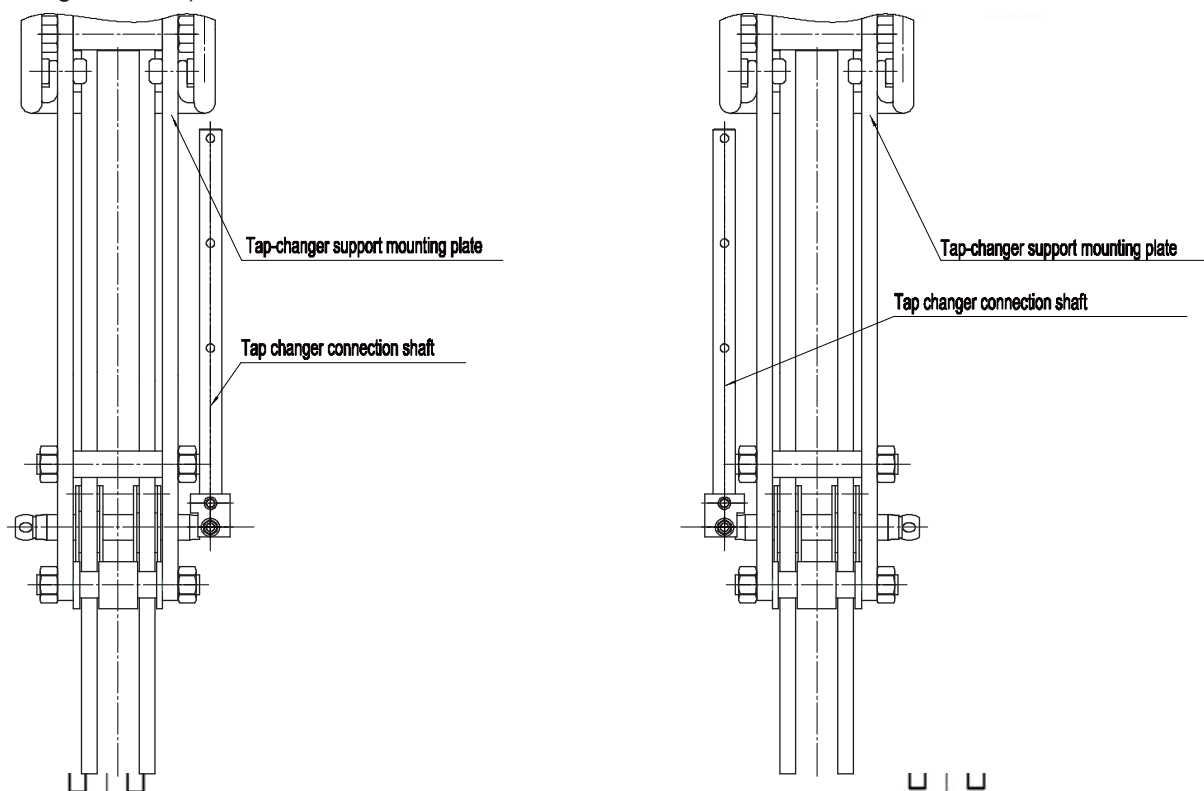


Before the drive shaft is installed and connected, please disassemble the relevant positioning part in advance.

2. Before installing the drive shaft, please check the deviations between tap-changers and the center line of the drive mechanism mounting flange, which cannot be greater than 6 degrees. (as is shown below)

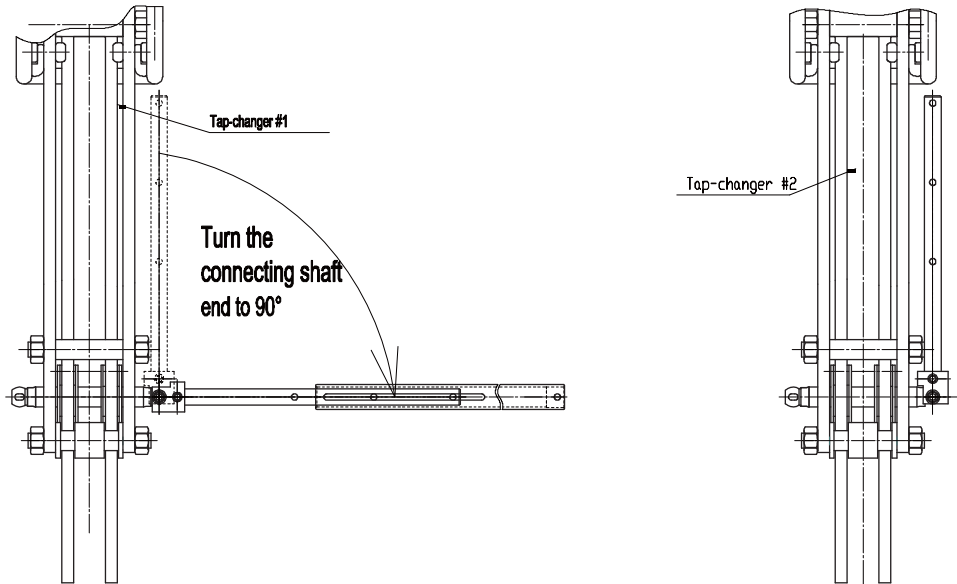


3. The tap-changer coupling shafts are assembled on the side of the support mounting plate when the tap-changer is shipped out (as shown in the left figure below). According to the transformer design requirements, the coupling shafts can be changed to the other side during on-site installation (as shown in the left figure below).

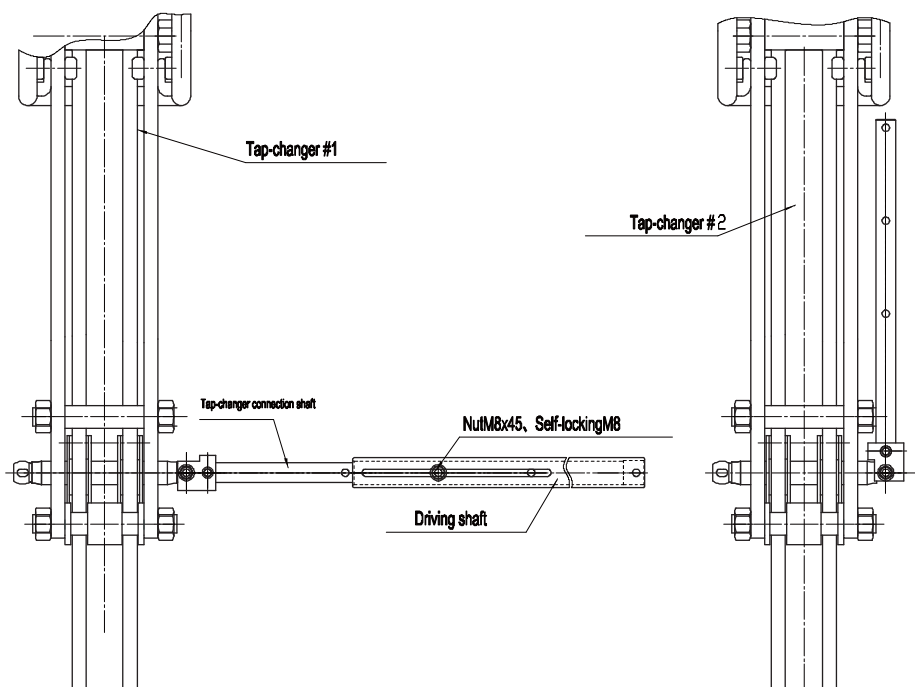


8.2.1 Drive shafts assembly between two tap-changers

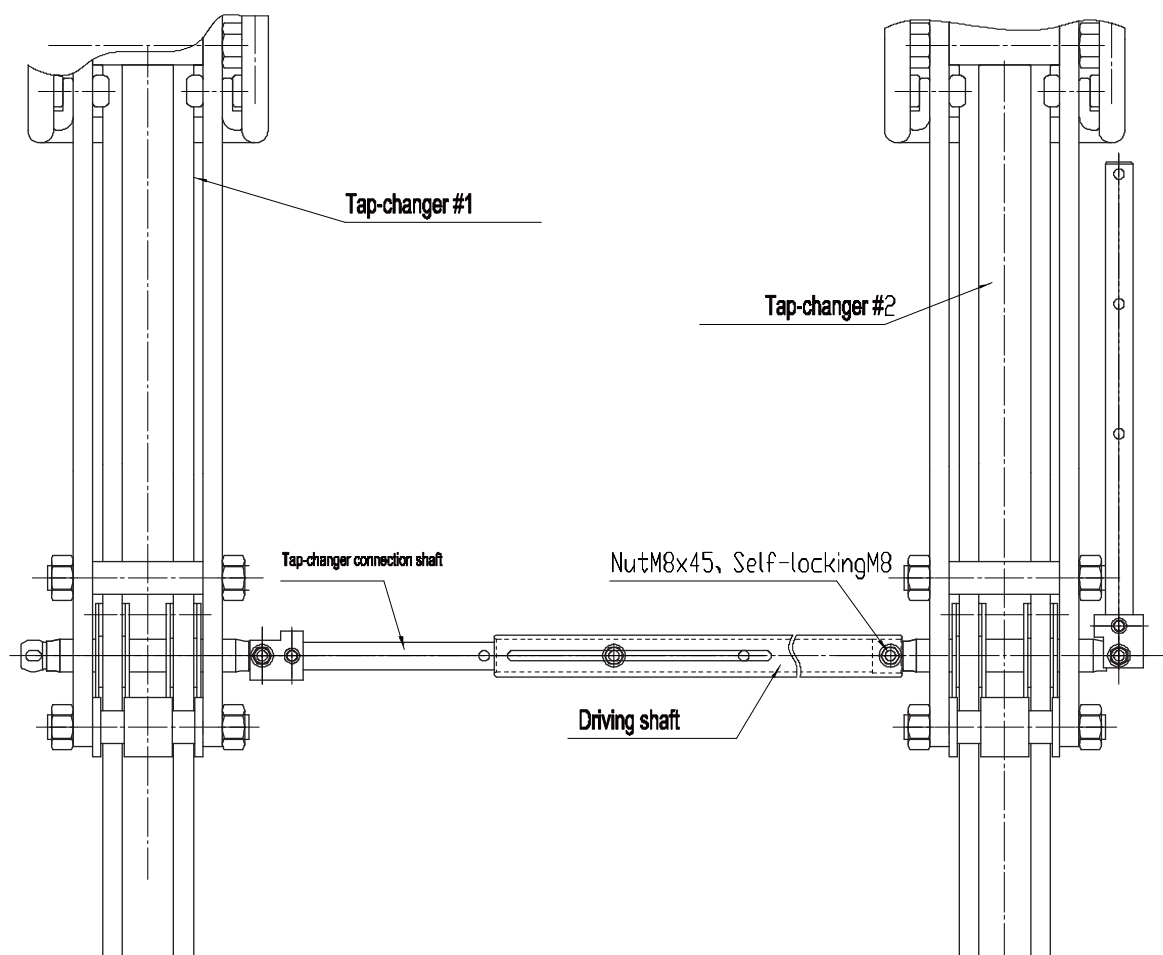
Step one: As is shown below, rotate the tap-changer coupling shafts 90 degrees.



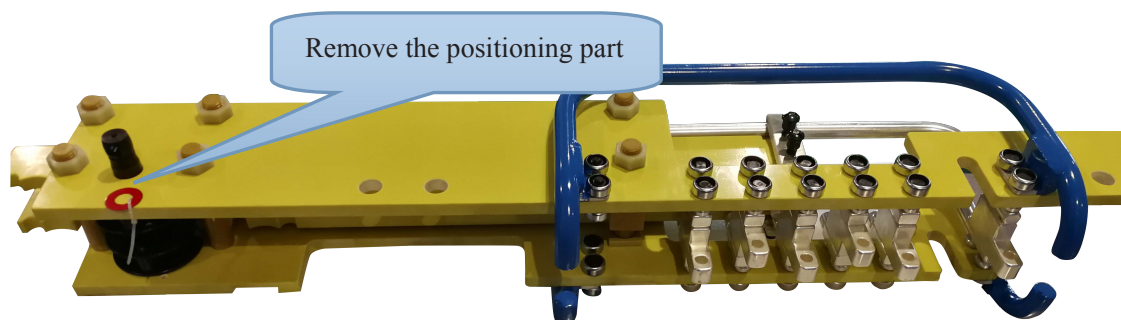
Step two: As is shown in the figure below, select the drive shaft of the corresponding length at the position according to the transformer design drawing (or the drawing confirmed by the transformer factory and the tap-changer manufacturer), and insert the drive shaft and the tap-changer coupling shaft. Then, the M8×45 screw is assembled from the screw hole of the drive shaft waist groove to the proper position of the coupling shaft, and the M8 self-locking nut is fastened (after fastening, the drive shaft and the tap-changer coupling shaft can move relatively to each other).



Step three: As shown in the figure below, align the drive shaft installed in the previous step with the shaft end of the other tap-changer and insert it. Then fasten with M8×45 screws and M8 self-locking nut.

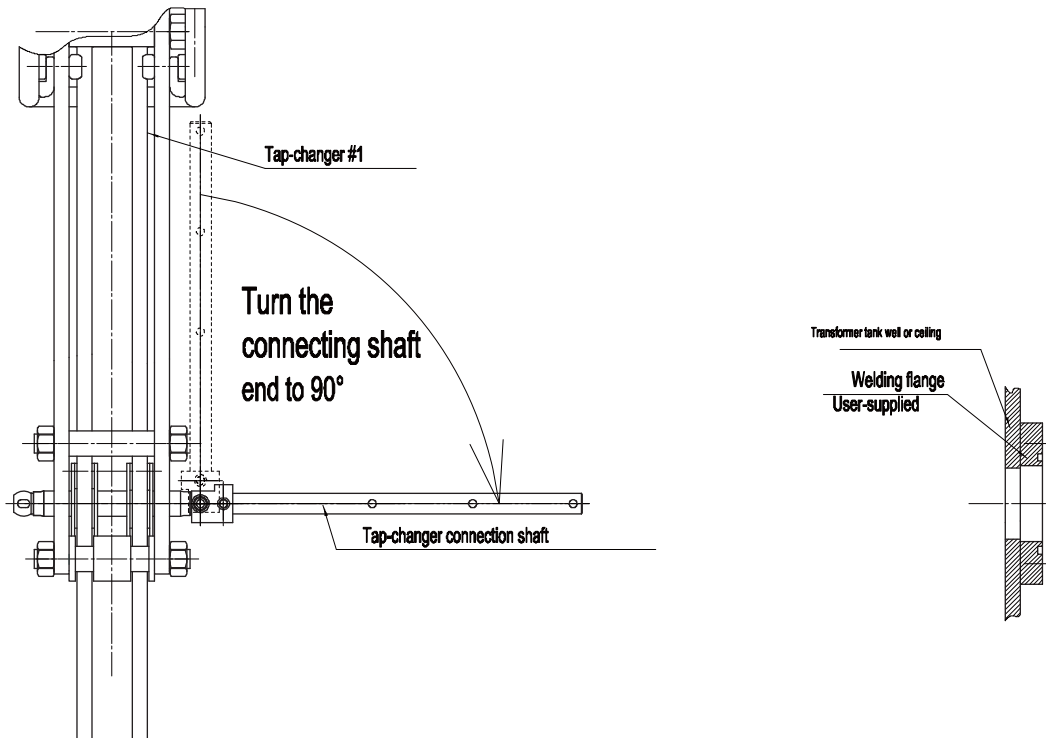


Step four: Remove the positioning part on the tap-changer as shown below.

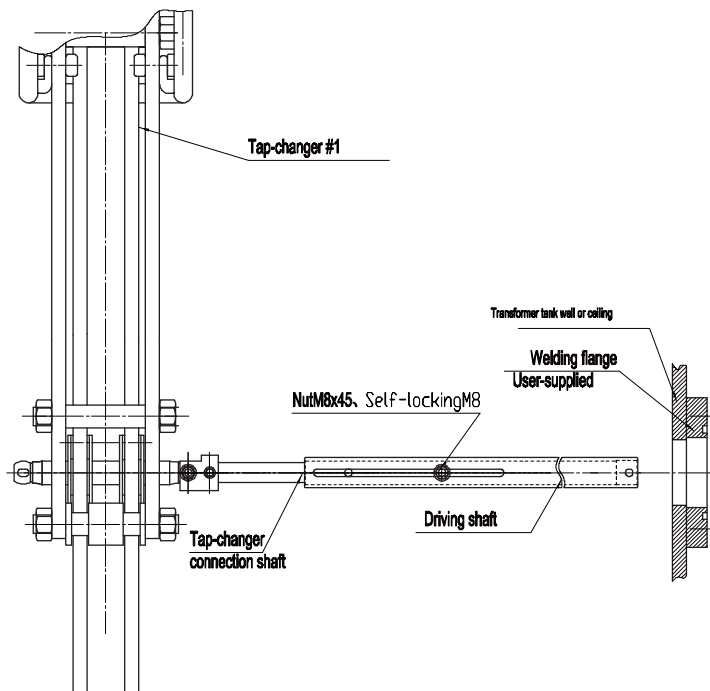


8.2.2 Assembly of the drive shaft between the tap-changer and the drive mechanism

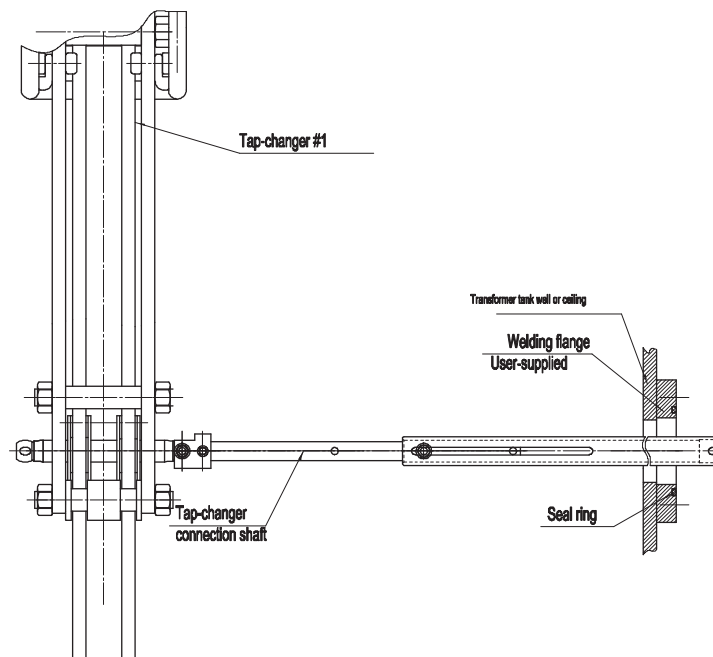
Step one: Rotate the tap-changer coupling shaft 90 degrees as shown below.



Step two: As is shown in the figure below, select the drive shaft of the corresponding length at the position according to the transformer design drawing (or the drawing confirmed by the transformer factory and the tap-changer manufacturer), and insert the drive shaft and the tap-changer coupling shaft. Then, the M8×45 screw is assembled from the screw hole of the drive shaft waist groove to the proper position of the coupling shaft, and the M8 self-locking nut is fastened (after fastening, the drive shaft and the tap-changer coupling shaft can move relatively to each other)

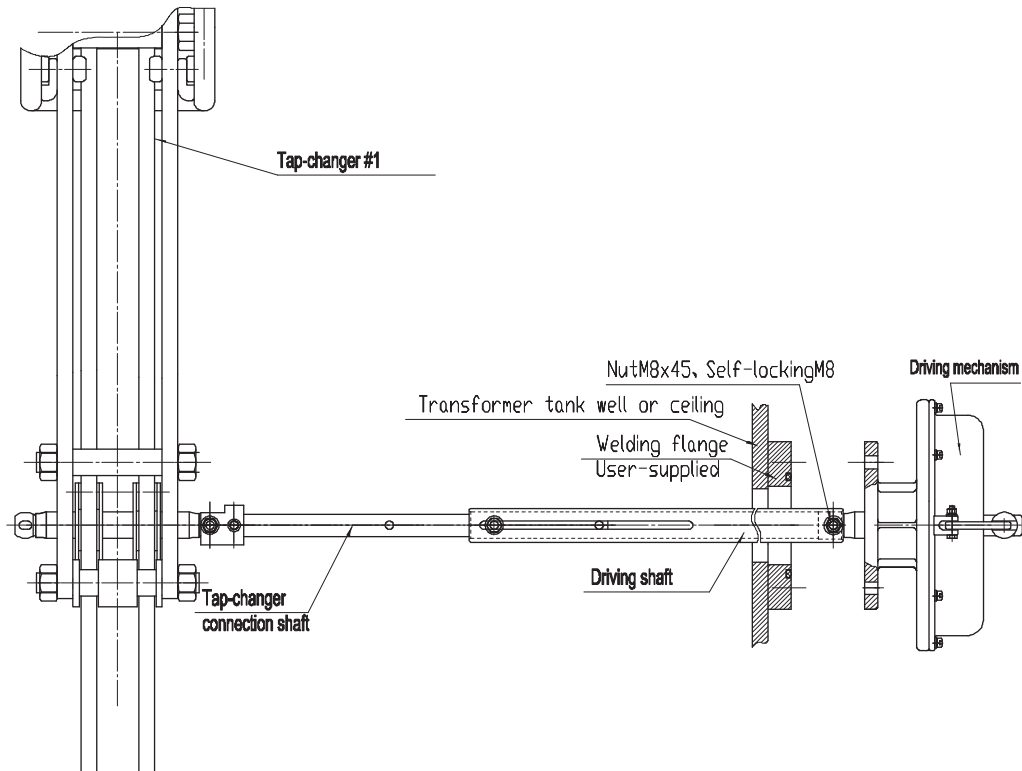


Step three: As shown on the figure below, pull out the drive shaft from the welding flange of the transformer and install the sealing gasket in the seal groove of the transformer installation flange, taking care not to rotate the drive shaft.

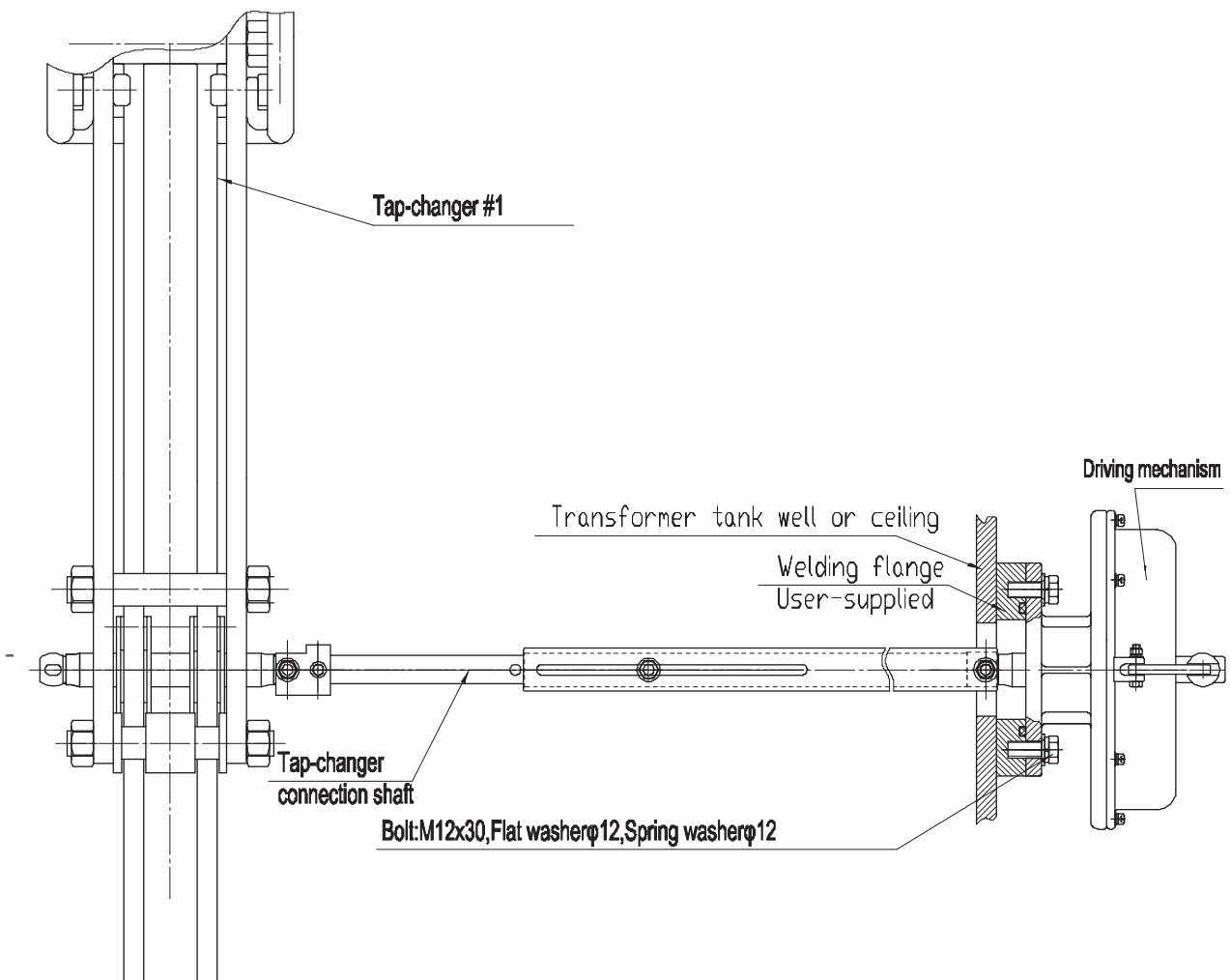


As shown in the figure below, align the drive shaft installed in the previous step with the shaft end of the other tap-changer and insert it. Then fasten with M8×45 screws and M8 self-locking nut, taking care not to

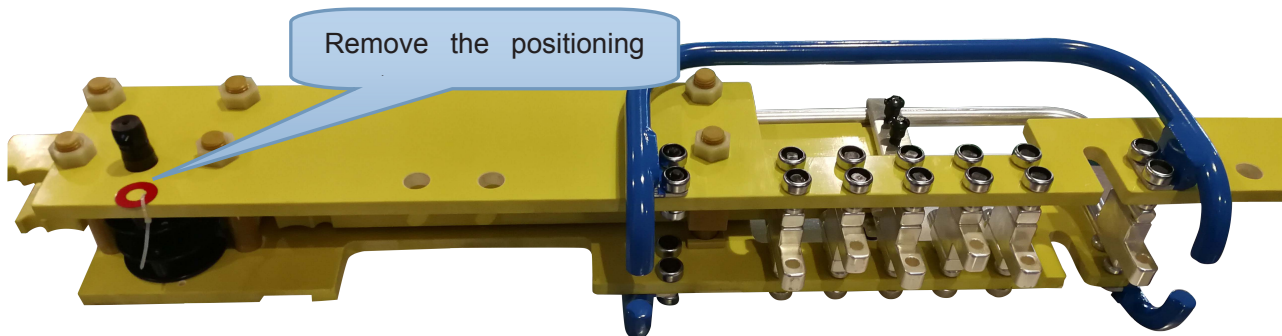
rotate the drive shaft.



Step five: As shown on the figure below, put the drive mechanism and the drive shaft close to the transformer mounting flange and place them in the positive position according to the transformer design drawings. Use M12x30 bolts, $\Phi 12$ flat washers and $\Phi 12$ spring washers provided by the tap-changer manufacturer to fasten. Be careful not to rotate the drive shaft.



Step six: Remove the positioning part on the tap-changer (if the transformer is of bell-type, this step can be advanced to after the second step and before lifting the transformer cover), as is shown below.



9. Transformer Ratio and DC Resistance Test

Before the transformer test, it is recommended to make the ratio test and the DC resistance test, and use the drive mechanism to switch positions to do the tests.

Switching from one working position to another, for full operation, it takes 1 to 4 cycles. Check whether each working position (position indicator on the tap-changer head cover) and the actual connecting position match or not. Measure the DC resistances of each position on both the step up and step down directions to see if they are stable and correct.



1. Before measuring the DC resistance, it is recommended to operate the tap-changer for at least 5 cycles!!!!
2. After the inspection, set the tap-changer to the middle position!!

10. Drying and Oil Filling

In order to ensure the insulation level, all tap-changers should be dried. Vacuum drying and vapor phase drying are both optional.

10.1 Vacuum drying

The tap-changer is put in the drying room at a temperature of about 60°C and is heated to 110°C at a rate of 10°C per hour in atmospheric pressure air. Then the tap-changer is pre-dried in a circulating hot air at 110°C for 20 hours and later dried again vacuum at 110°C for 50 hours with the residual pressure up to 110Pa.

10.2 Vapor phase drying

In a kerosene vapor at about 90°C, the tap-changer is kept for 3 to 4 hours, and then heated to a maximum temperature of 125°C at a rate of 10°C per hour. The drying time depends on the time required to dry the transformer.



1. After being dried, it is forbidden to operate the tap-changer without oil.
2. Even if it is needed to operate the tap-changer, it must be done by adding transformer oil to all tap-changer sliding contact parts.
3. Check if the fasteners are loose or not. If so, they must be retightened.
4. After installation, inject new transformer oil into the transformer. The oil quality must comply with ASTM standards

11.Operation and Maintenance



Make sure to operate and maintain the tap-changer while the transformer is de-energized!

It is strictly forbidden to use the tap-changer if the malfunction is not resolved!

For tap-changers with protective contacts, the protective contacts must be connected reliably!

Only when the installation, operation and commissioning are correct can the tap-changer be accepted and put into use. When the transformer is installed on site, the end-user or technicians of the transformer manufacturer should check the tap-changer operation status at first, and measure the transformer ratio and the DC resistance.

11.1 Tap-changer switching operation

- ① Every time after switching positions, DC resistance must be measured before being used again.
- ② Under normal circumstances, the operation torques of each position are basically the same. The larger the rated current of the tap-changer, the larger the shifting torque is.
- ③ The tap-changer is mechanically designed so that it can not go past its maximum position.

11.2 Maintenance

The following maintenance should be performed at least every year or as needed during normal operation.

- ① Perform at least two full-range tapping cycles on the tap-changer. Perform DC resistance test on each

position and confirm that it is normal before continuing to use it.

- ② Clean the dust on the tap-changer, and rust-proof the rusted parts or the whole product. Pay attention to keep the position indicator correct and clear.

When disassembling the drive mechanism, mark the mounting flange and the welding flange for reloading; non-professionals should not disassemble the tap-changer internal components. If in doubt, contact the transformer factory or the tap-changer manufacturer in time.

12. Troubleshooting

12.1 Leakage

1. If the area between the tap-changer mounting flange and the welded flange has oil leakage: tighten the fixing bolts of the mounting flange and replace the sealing gasket.
2. If the shaft seal between the tap-changer main shaft and the mounting flange has oil leakage: remove the head drive mechanism, insert an oil seal to the input shaft of the head drive mechanism and re-assemble the removed parts.



Mark the positions, sequences and numbers of components before disassembly to avoid mistakes or missing parts when doing reassembly; record the original indication position to avoid wrong position.

12.2 Abnormal DC resistance

1. If the tap-changer is not installed and verified as required, the DC resistance may be abnormal.



Installation and verification of the installation must be carried out in strict accordance with the instructions. Otherwise failure may occur even if the DC resistance is normal.

2. The measured ratio and the DC resistance do not match the order of the required values, indicating that the tap-changer position is inconsistent with that of the drive mechanism. For example, the mechanism is set at Position 3 but the tap-changer is actually at Position 2). This problem is because when the tap-changer is re-assembled, it is not set at the original position.

Solution: Remove the drive mechanism, rotate the shaft end of the mounting flange, make the red line mark of the shaft end align with the red line mark of the position number of the installation flange so that the position number is the same as that of the drive mechanism, and the re-assemble position

number is the same as that of the drive mechanism, and the re-assemble the components. .



Mark the positions, sequences and numbers of components before disassembly to avoid mistakes or missing parts when doing reassembly; record the original indication position to avoid wrong position.

3.The DC resistance of the same position is unstable. It may be that the contact pressure is asymmetrical or the side with lower pressure is heated by long-term current and oxidized, resulting in malfunction.

Solution: Adjust the tap-changer and drive mechanism so they can synchronize or re-install as required, eliminate installation deviation, and then operate the positions at least 5 full-range cycles to wear off the oxide layer. Measure the DC resistance again. If the contact is found to be discolored or slightly ablated, they should be sanded. In severe cases, the damaged parts or the whole tap-changer must be replaced.

4. For tap changer that have not been energized or operated for a long time, a very thin oxide film may form on the contact surfaces.

Solution: Perform at least 5 full-range cycles to remove the oxide film and measure the DC resistance again.

5.After operation, slight change of the surface of silver-plated contacts is a normal phenomenon. If there is a large area of black stain and the DC resistance is unstable and irregular, it means sulfur content in the oil may exceed the standard.

Solution: A comprehensive test should be done. The contacts should be processed or replaced and the tap-changer should be replaced if necessary.

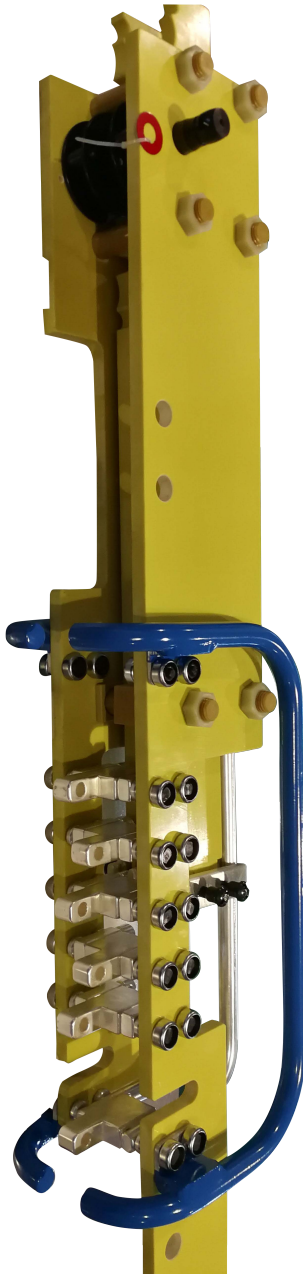
13.Warranty

Free repairs are provided if the product fails to function properly due to manufacturing quality problems within 12 months from the date of installation or within 18 months from the date of shipment to the transformer manufacturer.

If the tap-changer or drive mechanism has a serious failure and it is not easy to repair on site, please contact the after-sales service department directly.

14.Appendix

14.1 Tap-changer pictures



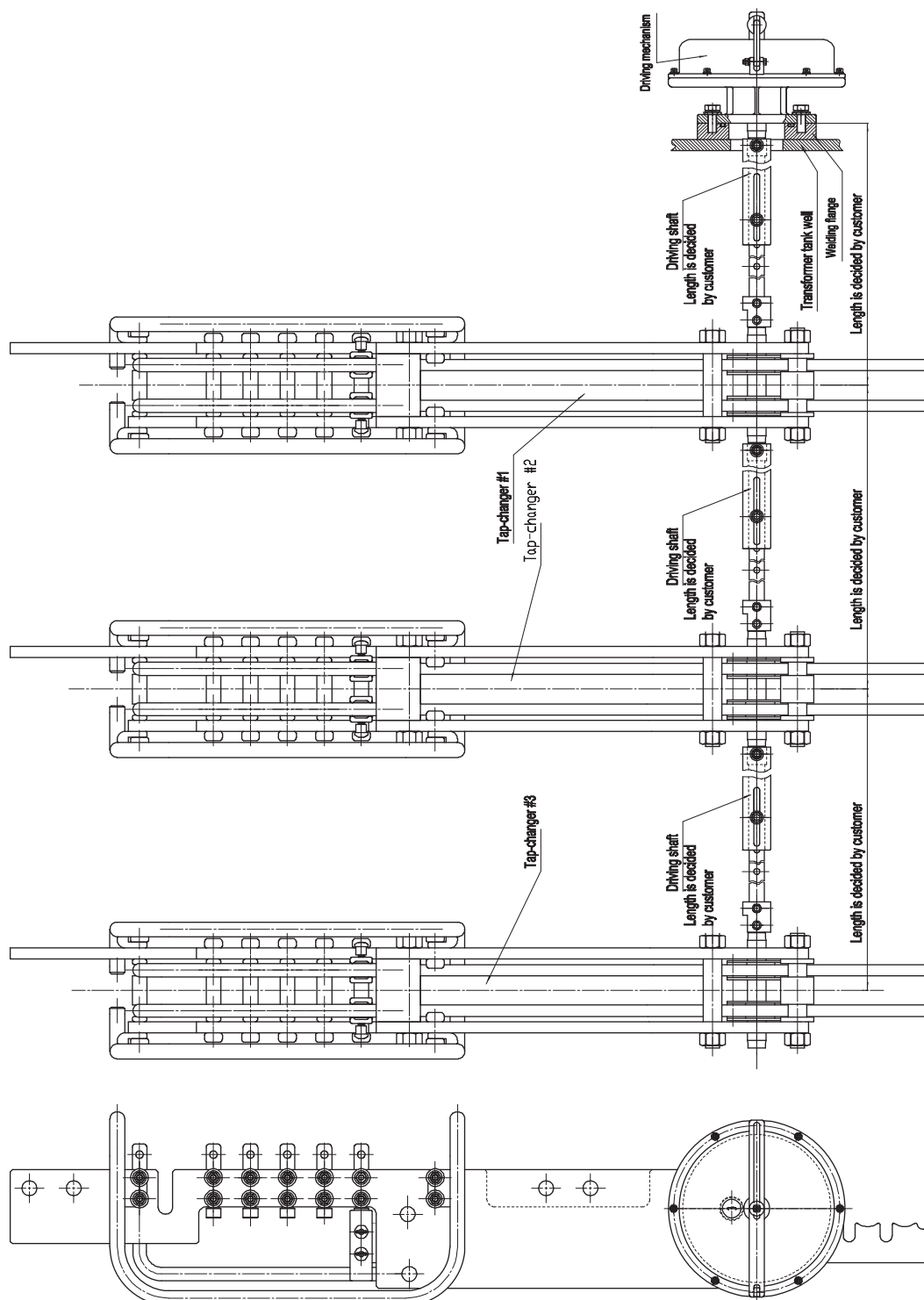
With shielding 1000A



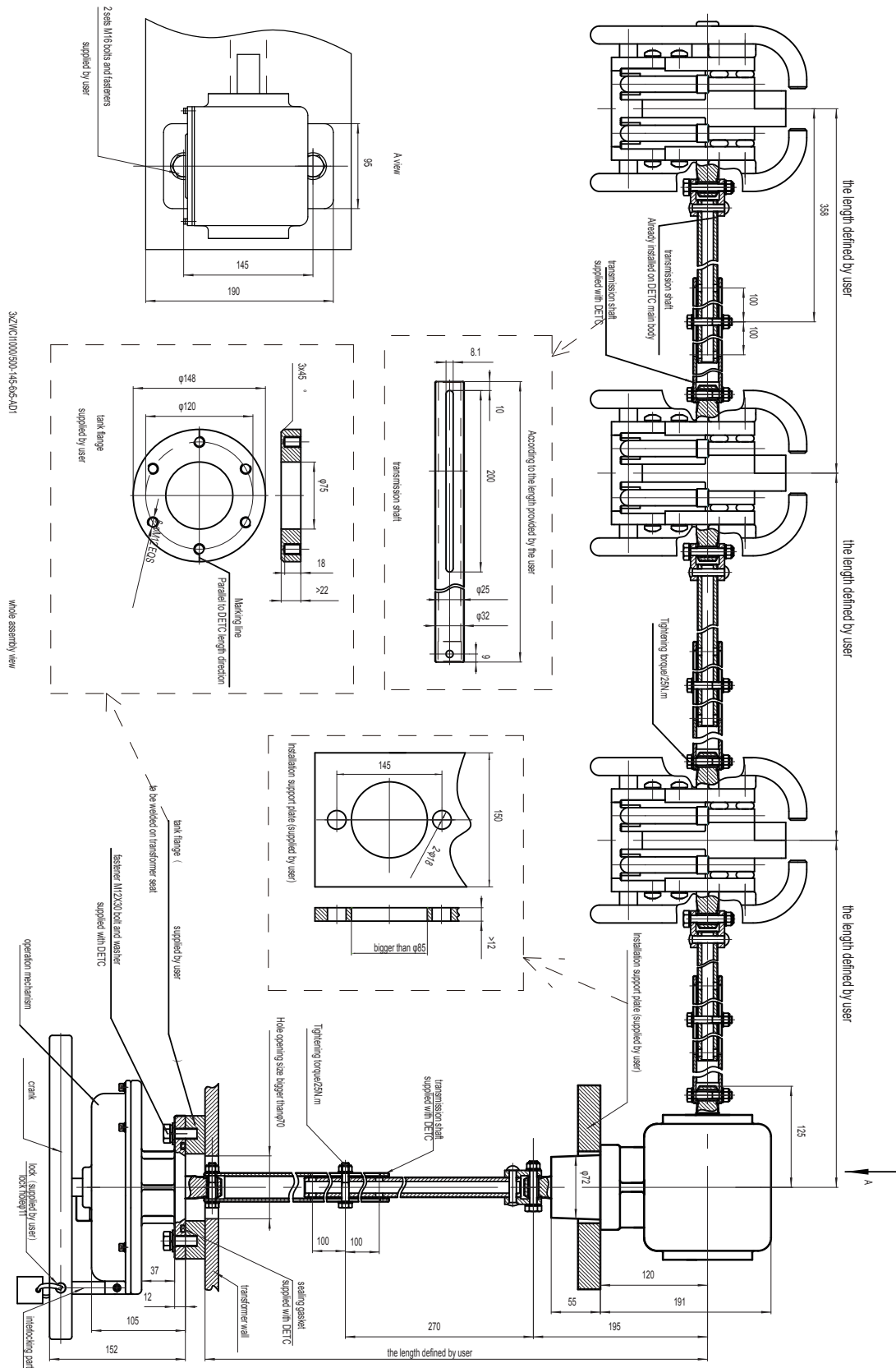
Without shielding 1000A



14.6 Three phase assembly of tap-changer



14.7 Assembly with angle gear configuration



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